DOCKHUB · PROJECT PLAN

DockHub project plan

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	DKH-PLN-001	Version	0.1
Status	Draft	Date	2026-10-31
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dockhub		
Source	docs/11-plan.md @ 5167e03		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	First plan at the kit 1.9.1 upgrade; stages P0 to P3 done on paper, P4 to P6 on hold under the stage cap

DockHub project plan

DockHub is at stage P3 (design outputs), TRL 3. The current design is option A; the concept choice is proposed, awaiting Amish (DKH-DDR-004). The calculation note, the D3 model of all three options, the drawings, the priced BOM and the build plan are done. The next gate, P4 (verification, TRL 4), is on hold while the portfolio stage cap is P3 (`.kit/PHASE.yaml`). Method: STANDARDS section 19.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept choice	Amish Chadha	DKH-DDR-004, DKH-OPT-001	Proposed, awaiting Amish
Partner and site	Amish Chadha	A delivery-worker group, New York City first, on private property (DKH-DEC-001)	Not started
Requirements	Design lead	DKH-URS-001, DKH-PRS-001, DKH-RTM-001	Done on paper; partner review pending
Calculations and model	Design lead	DKH-CAL-001; D3 model of options A, B and C; STEP and STL	Done 2026-10-31
Drawings, BOM, build plan	Design lead	DKH-DWG-001, DWG-101 onward, bom/bom.csv, DKH-BLD-001	Done
Risk, market, regulatory	Design lead	DKH-RSK-001 with design FMEA, DKH-MKT-001, DKH-REG-001	Done 2026-10-31
Renders	Coordinator	Options picture (Blender), product renders	Options picture to render by the coordinator
Verification (P4)	Partner workshop	Propagation trial, detector timing, IP54 test, build	On hold (stage cap P3)

Resources needed

Table 2. Resources.

Resource	Deliverable	Effort	Estimated cost (USD)
Design lead (Claude with Amish's review)	P1 to P3 documents	Sessions done	0
Prototype parts	Two-bay cabinet with anchors and pad (P4, on hold)		1,519 (bom/bom.csv; target 1,200)
SwapCell packs	One resident pack for the prototype		414 each (SwapCell BOM, not in the cabinet cost)
Sheet metal shop and electrician	Cabinet and mains connection		To assess
Fire certification route	UL 1487 test and FDNY approval (DKH-REG-001 G1, G2)		To assess; FDNY lists a USD 625 application fee

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Done 2026-10-31
P1 Definition	URS, PRS, RSK, MKT, REG complete	Done 2026-10-31
P2 Design inputs	PRS targets, CAL, RTM, DDR, STEM check closed by calculation	Done 2026-10-31; ten STEM rows need hardware
P3 Design outputs	Model (D3), STEP, DWG, BOM, BLD, DEC, FMEA	Done 2026-10-31
P4 Verification	Propagation trial, detector timing, enclosure test, charge pause on a hot day	On hold
P5 to P6	Pilot with a partner, launch	On hold

Next steps and decision requested

Requested from Amish: the concept choice (DKH-DDR-004, option A recommended), whether to keep the USD 1,200 target against a USD 1,519 estimate, and whether the fire rows S11, S13 and S23 may stay open as TRL 4 trial items. The recommended next step is a partner and, when the cap allows, TRL 4: the propagation trial with plain and vermiculite bay floors.

References

Gogoro. "Gogoro Network Flex Plan." September 19, 2019. https://www.gogoro.com/news/gogoro-network-flex-plan/.

CleanTechnica. "Battery Swapping Expands In New York City." February 28, 2026. https://cleantechnica.com/2026/02/28/battery-swapping-expands-in-new-york-city/.

FDNY. *E-Micromobility Approval Guide.* Accessed October 31, 2026. https://www.nyc.gov/assets/fdny/downloads/pdf/business/e-micromobility-approval-guide.pdf.

Levy Electric. "Swobbee Battery Swapping Station" and "Gogoro GoStation." Accessed October 31, 2026. https://fleets.levyelectric.com/docking-stations/swobbee-battery-swapping-station; https://fleets.levyelectric.com/docking-stations/gogoro-gostation.

NYC Department of Transportation. Press release, November 18, 2025. https://www.nyc.gov/html/dot/html/pr2025/install-public-e-bike-battery-swapping-network-citywide.shtml.

US Access Board. "Chapter 3: Operable Parts." Accessed October 31, 2026. https://www.access-board.gov/ada/guides/chapter-3-operable-parts/.